

COST OF OPPORTUNITY PROBLEM SET

ECON101—INTRODUCTION TO MICROECONOMICS.

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Problem 1

Suppose that University Service Management (USM) is considering whether to spend 9.5 billion in building a new recreation center/food court. All of NU students agree that the existing options on campus for both food and entertainment leave much to be desired.

Despite this, when students are asked to vote on whether they would like USM to build the new center, over 85% voted no. Why do you think that such a large fraction of students voted no even though all of them agree that a new recreation center/food court would be much nicer than what they currently have?

Problem 2

Beksultan describes himself as the “Gigachad of the Finance Bros”. His favorite movie is the *Wolf of Wall Street*, he casually inserts phrases like “I’m really into Finance because I have such a passion for the markets” into all conversations, and he is constantly checking stocks during the ECON101 class, even though he has no idea what any of the numbers mean.

The “only” problem in Beksultan’s mind is that he doesn’t have the outfit that would signal to everyone his prowess as a Finance Bro. You know the outfit: beige chinos, branded t-shirt, quarter-zip pullover sweater, leather loafers, recognizable-brand watch, leather backpack, and Ray-Ban glasses. The total cost of this outfit is approximately half a million tenge. The benefit of having the outfit is proportional to the number of people Beksultan impresses with it; but let’s be honest, nobody cares except his closest circle of Finance Bros. In fact, many people outside the Finance Bros would find such behavior loathsome and would actually try to avoid Beksultan because of it, i.e., it would be an *implicit cost*.

Given the information above, what is the correct way to calculate Beksultan’s surplus?

Problem 3

Consider the same setup as the previous problem. Beksultan made the mistake of disclosing his intention to buy the outfit to all of his acquaintances in the Finance Bro circle. Consequently, all of the Finance Bros went ahead and bought similar outfits before Beksultan. The problem is that a person cannot impress anyone in their social circle if all of them are similar or homogeneous. After seeing this, but before buying his own outfit, Beksultan ponders what to do. What would you recommend he do in this scenario, and why?

Problem 4

During the summer, Kamilla grows tomatoes in her grandfather’s garden. She then sells them at the local farmer’s market for 1500 tenge per kilogram. By adding compost to the garden, she can increase

the yield as shown in the table below. If compost costs 1850 tenge per kilogram and Kamilla's goal is to make as much profit as possible, how many kilograms of compost should she add?

Kilograms of compost	Kilograms of tomatoes
0	100
1	120
2	125
3	128
4	130
5	131
6	131.5

Problem 5

Daniyar is the best mushroom farmer in town. He invests all of his spare cash in additional mushrooms, which he grows on otherwise useless land behind his barn. The mushrooms double in weight during their first year, after which time they are harvested and sold at a constant price per kilogram. Daniyar's friend Bagdat asks Daniyar for a loan of 200,000 tenge, which he promises to pay after one year. How much interest will Bagdat have to pay Daniyar in order for Daniyar to recover his opportunity cost of making the loan?

Problem 6

Suppose that in the last few seconds you devoted to question 1 of your Economics exam you earned 4 extra points. while in the last few seconds you devoted to question 2 you earned 10 extra points. You earned a total of 48 and 12 points, respectively, on the two questions, and the total time you spent on each was the same. If you could take the exam again, how—if at all—should you reallocate your time between these questions?

Problem 7

Akmaral and Medina have the same preferences and income. Just as Akmaral arrived at the cinema to see a movie, she discovered that she had lost the 1,600 tenge ticket she had purchased earlier. Medina also just arrived at the cinema planning to buy a ticket to see the same movie when she discovered that she had lost 1,600 tenge from her wallet. If both Akmaral and Medina are rational and both still have enough money to pay for a ticket, is one of them more likely than the other to go ahead and see the movie anyway?

Problem 8

Suppose there is only one electric scooter company in Astana. Currently, the company charges 200 tenge per minute, and there is no fee to unlock the scooter. The scooter company is considering changing its pricing plan so that it would charge 1000 to unlock the scooter and 100 tenge per minute. If Nazima never takes a ride that lasts less than 10 minutes, then what will happen to the average length of her rides if the scooter company switches to the new pricing plan?

Problem 9

Suppose that NU is trying to implement a meal plan for its students. There are two proposals, which we will call plan A and plan B. Plan A would let students eat as much as they want in the university's canteens for a fixed fee of 250,000 tenge per semester. NU estimates that the average student consumes around 100 kg of food per semester. Plan B charges 250,000 tenge for a book of meal tickets that entitles the student to eat 100 kg of food per semester. If the student eats more than 100 kg, he or she pays 2,000 tenge for each additional kilogram; if the student eats less, he or she gets a 2,000 per kilogram refund. If the students are rational, under which plan would the average food consumption be higher?